

Week 16

Advanced Data Ethics, Privacy, and Responsible AI

North Metropolitan TAFE

Certificate IV in Information Technology (Data Science and AI)

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Location. East Perth Campus

Learning Objectives for Week 16

- Understand higher-level ethical considerations when working with large, open datasets for AI.
- Analyze issues of consent and fair representation impacting modern machine learning.
- Evaluate data privacy risks and privacy-preserving techniques in real-world AI workflows.
- Assess how changing policy and legal requirements affect dataset engineering for AI.

This Week at a Glance

- Review. Building on machine learning and large dataset skills from Weeks 13 to 15.
- Key Topics. Consent, fair representation, bias, privacy, and regulatory change in AI.
- Industry Lens. Real-world challenges faced by AI teams working with open data.
- Activities. Case analysis, ethics review, and privacy tool exploration.

Why Advanced Ethics Matters in Modern AI

- LLMs and generative models learn from massive, often messy datasets scraped from the internet.
- Ethical lapses can undermine trust, harm marginalized groups, or break laws.
- IT and data science teams must align with standards for ethical curation, transparency, and responsible AI.

Consent and Fair Representation in AI Datasets

- Many large datasets lack explicit consent from data subjects; this raises ethical and legal questions.
- Fair representation means ensuring diverse populations are not excluded, misrepresented, or stereotyped.
- Open datasets like LAION-5B and COCO highlight the complexity of tracking consent and demographic equity.

Dataset Bias. Real-World Consequences

- AI models trained on biased data can amplify stereotypes or systematically disadvantage groups.
- Example. Image datasets overrepresent certain regions, genders, or professions. Text datasets repeat cultural biases.
- Industry Impact. Failing to address bias can lead to flawed products, reputational harm, and regulatory scrutiny.

Privacy-Preserving Methods in AI and ML

- Common strategies include data anonymization, pseudonymization, differential privacy, and federated learning.
- Tools such as HuggingFace Datasets and industry ML libraries integrate some privacy features.
- Organizations must balance data utility for AI against risks to individuals' privacy.

Responding to Evolving Policy and Regulation

- Laws such as GDPR and emerging AI Act require active management of consent, use, and documentation.
- Open dataset maintainers must track data sources, permissions, and compliance.
- AI practitioners need workflows for ethical impact assessment and regular policy review.

Industry Case Study. Ethical Failure in AI Dataset Curation

- A facial recognition project trained on publicly scraped images lacked proper consent, resulting in privacy violations and lawsuits.
- Lack of documentation and bias in the dataset led to both technical and ethical failures.
- Solution involved re-auditing the dataset, removing problematic data, increasing transparency, and implementing more responsible sourcing practices.

Hands-On Activities This Week

- Ethics Audit. Review an open dataset's documentation for consent, representation, and privacy practices.
- Bias Analysis. Identify examples of under- or over-representation in a real dataset using Python or pandas.
- Privacy Tool Demo. Explore differential privacy with available methods in open-source ML libraries.
- Group Debate. Should all public web data be fair game for AI training? Discuss limits and risks.

Assessment Preparation

- Upcoming Report. Link findings from ethics and privacy reviews to the research report assessment.
- Test Readiness. Practice identifying ethical risks and proposing mitigation strategies for dataset scenarios.
- Review Key Concepts. Consent, bias, privacy, and policy for knowledge-based assessment (Week 18).

Summary and What's Next

- Advanced data ethics, privacy, and responsible AI are essential for safe and effective ML projects.
- This week's skills will support you in completing the research report, ML project, and knowledge assessment.
- Next Week. Synthesis phase; integrating skills to finalize open dataset and ML projects; focus on practical demonstration of ethical awareness.

